File System Implementation

Understanding Operating System Storage Architecture

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Layered Architecture

The file system is implemented as a vertical stack of layers, with each layer building on the one below it.

Application Programs

Logical File System

File Organization Module

Basic File System

I/O Control

Devices

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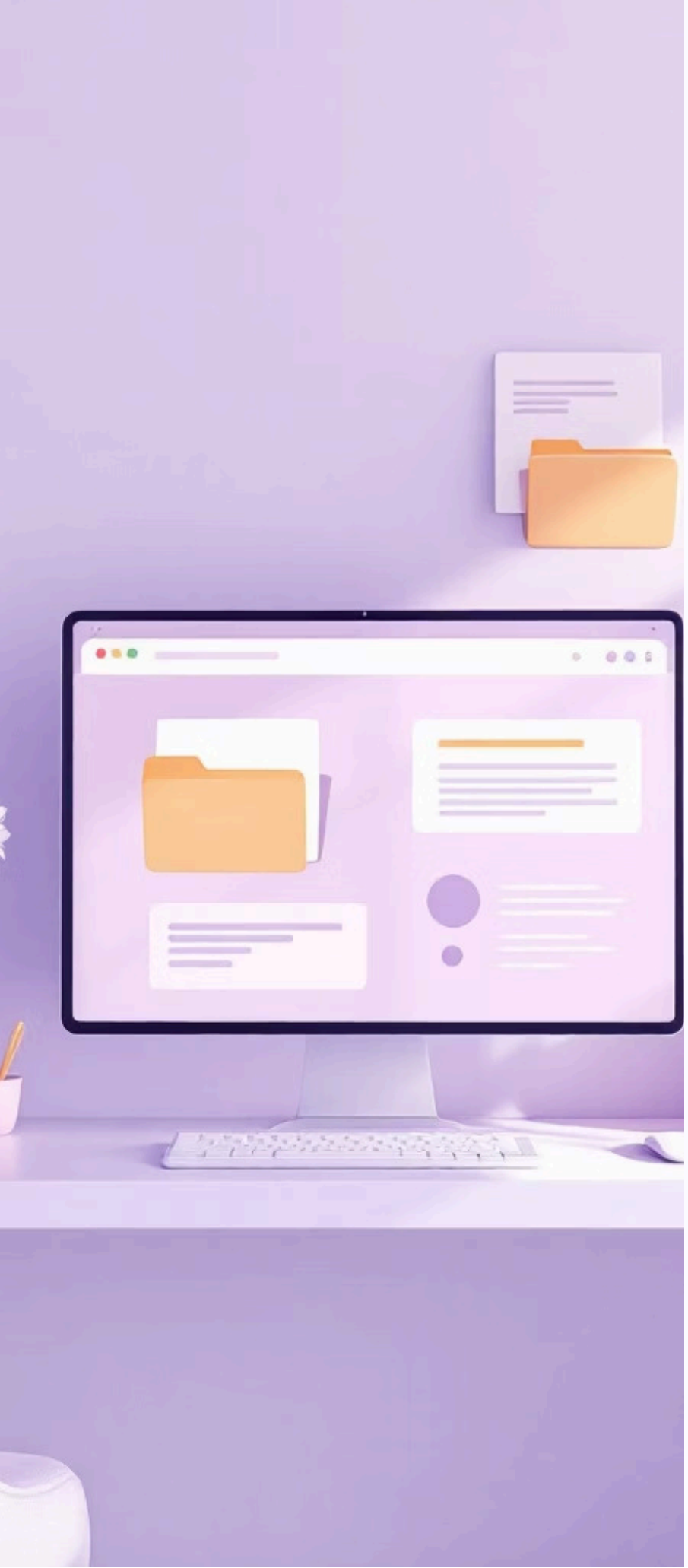

Detailed Layer Explanations

Application Programs

User-facing apps that request file operations through system calls.

Logical File System

Manages metadata, directories, permissions, and file control blocks.

File Organization Module

Translates logical file blocks into physical disk blocks and tracks free space.

Basic File System

Handles block allocation, buffering, and free-space management.

I/O Control

Coordinates device drivers and manages low-level I/O requests.

Devices

Physical storage hardware that reads and writes data.

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Two Types of Data Structures

On-Disk Structures

Stored permanently on disk storage. Include boot control block, volume control block, directory structure, and file control block.

In-Memory Structures

Stored temporarily in RAM. Include mount tables, directory cache, system-wide open file tables, and per-process open file tables.

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On-Disk Structures

Permanent structures that store critical information about files and disk layout.

01

Boot Control Block

First block of volume containing boot information. Called boot block in UNIX and partition boot sector in NTFS.

02

Volume Control Block

Information about partition including free block count, block size, and pointers. Superblock in UNIX, stored in master file table in NTFS.

03

Directory Structure

Stores file names and associated inode numbers. File names and inodes in UNIX, stored in master file table in NTFS.

04

File Control Block (FCB)

Contains file details with unique identifier for directory association. Stored in master file table in NTFS.

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File Control Block Structure

File Permissions

Access rights and security attributes

File Dates

Create, access, and write timestamps

Owner & Group

File owner, group, and ACL information

File Size

Total size in bytes

Data Blocks

Pointers to actual file data blocks

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In-Memory Structures

Temporary structures maintained in RAM for efficient file system management and caching.

Mount Table

Contains information about each mounted volume and file system.

Directory Cache

Holds directory information of recently accessed directories for faster lookup.

System-Wide Open File Table

Contains copy of FCB for each open file across all processes.

Per-Process Open File Table

Tracks files opened by each process and maps to system-wide table.

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Key Takeaways

Layered Design

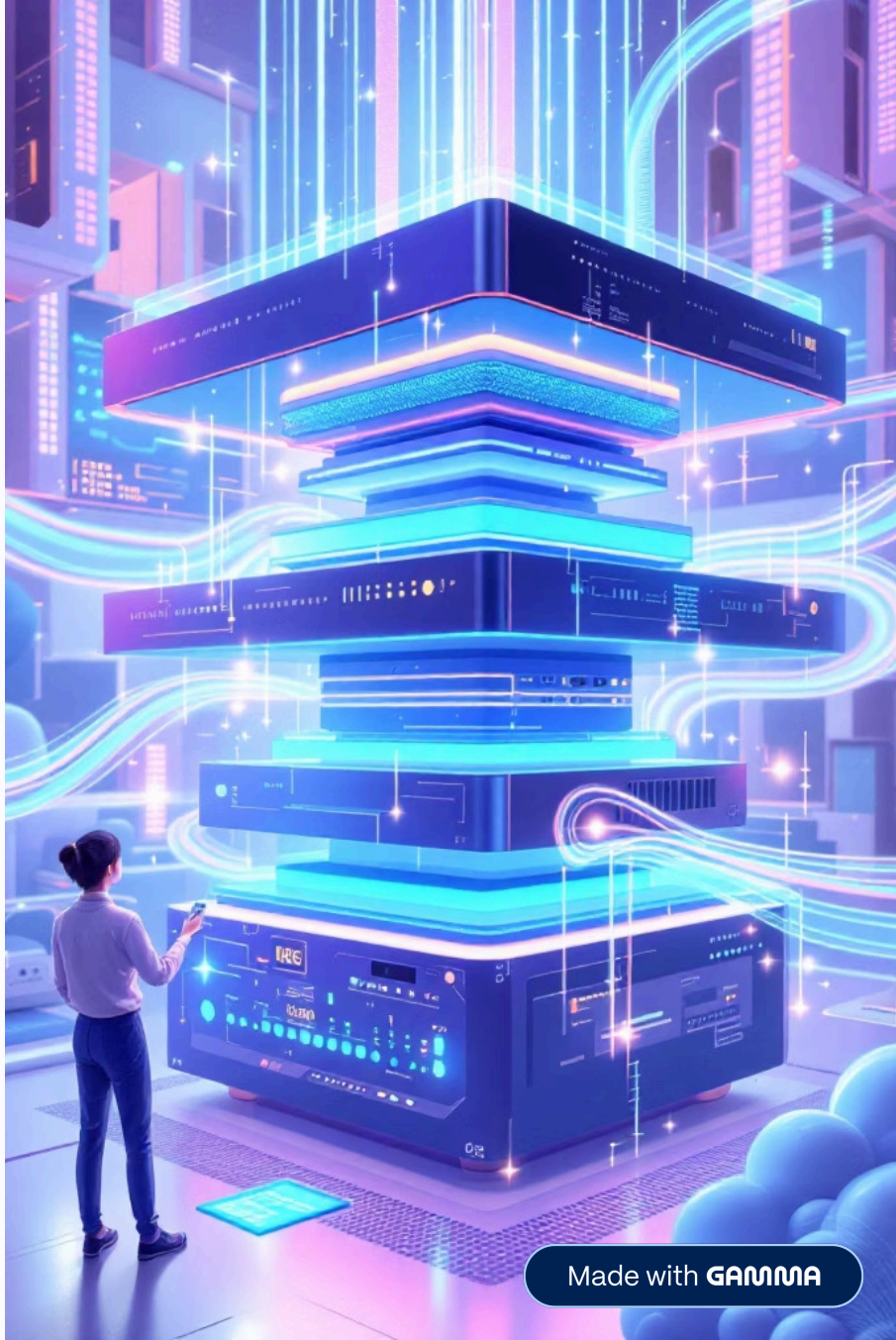
Each layer provides abstraction and specific functionality for efficient file management.

Two Structure Types

On-disk structures store permanent metadata while in-memory structures enable fast access.

Efficient Access

Caching and table structures minimize disk access and improve system performance.



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Thank You

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